

2.7.28

EE25BTECH11016 - Taraka Abhinav

Question:

Find the area of the triangle whose vertices are $A(2, 3)$, $B(3, 5)$, and $C(4, 4)$.

Solution:

The area of a triangle with vertices **A**, **B**, and **C** can be calculated using the determinant of the vectors forming two of its sides. The formula is:

$$\text{Area} = \frac{1}{2} |\det(\mathbf{A} - \mathbf{B}, \mathbf{A} - \mathbf{C})|$$

The given vertices are represented by 2D vectors:

$$\mathbf{A} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$$

First, we find the two side vectors:

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 5 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \end{pmatrix} \quad (1)$$

$$\mathbf{A} - \mathbf{C} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 4 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \end{pmatrix} \quad (2)$$

Next, we calculate the determinant of the matrix formed by these two vectors as columns:

$$\begin{aligned} \det(\mathbf{A} - \mathbf{B}, \mathbf{A} - \mathbf{C}) &= \begin{vmatrix} -1 & -2 \\ -2 & -1 \end{vmatrix} \\ &= (-1)(-1) - (-2)(-2) \\ &= 1 - 4 = -3 \end{aligned} \quad (3)$$

The area is half the absolute value of this determinant:

$$\text{Area} = \frac{1}{2} |-3| = 1.5 \quad (4)$$

Thus, the area of the triangle is 1.5 square units.

